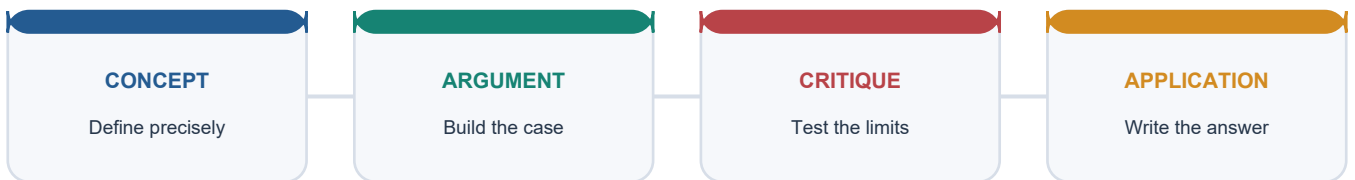


SOLVED PRACTICE WORKBOOK

Logical Reasoning Coding Counting and DI - CSAT Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Logical Reasoning Coding Counting and DI - Solved Practice Workbook 6

BASIC MCQS / REMEDIATION 6

Diagnostic set - Questions 1-16	6
Q1. Practice variant 1: Complete the series: 2, 4, 8, 14, ?	6
Q2. Practice variant 1: Each letter is shifted forward by 1. How is CAT coded?	6
Q3. Practice variant 1: A person faces North and turns right 0 time(s), each by 90 degrees. Which direction is faced?	6
Q4. Practice variant 1: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	6
Q5. Practice variant 1: All poets are readers. No reader is silent. Which conclusion follows?	6
Q6. Practice variant 1: How many ordered selections of 2 people can be made from 5 distinct people?	6
Q7. Practice variant 1: A bag has 2 red and 3 blue balls. One is drawn at random. Probability of red?	6
Q8. Practice variant 1: A table lists three values: 20, 30, 50. What is their total?	6
Q9. Practice variant 2: Complete the series: 3, 5, 9, 15, ?	6
Q10. Practice variant 2: Each letter is shifted forward by 2. How is CAT coded?	6
Q11. Practice variant 2: A person faces North and turns right 1 time(s), each by 90 degrees. Which direction is faced?	6
Q12. Practice variant 2: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	7
Q13. Practice variant 2: All poets are readers. No reader is silent. Which conclusion follows?	7
Q14. Practice variant 2: How many ordered selections of 2 people can be made from 6 distinct people?	7
Q15. Practice variant 2: A bag has 3 red and 3 blue balls. One is drawn at random. Probability of red?	7
Q16. Practice variant 2: A table lists three values: 21, 31, 51. What is their total?	7
Graded set - Questions 17-32	7
Q17. Practice variant 3: Complete the series: 4, 6, 10, 16, ?	7
Q18. Practice variant 3: Each letter is shifted forward by 3. How is CAT coded?	7
Q19. Practice variant 3: A person faces North and turns right 2 time(s), each by 90 degrees. Which direction is faced?	7
Q20. Practice variant 3: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	7
Q21. Practice variant 3: All poets are readers. No reader is silent. Which conclusion follows?	7
Q22. Practice variant 3: How many ordered selections of 2 people can be made from 5 distinct people?	7
Q23. Practice variant 3: A bag has 4 red and 3 blue balls. One is drawn at random. Probability of red?	7
Q24. Practice variant 3: A table lists three values: 22, 32, 52. What is their total?	7
Q25. Practice variant 4: Complete the series: 5, 7, 11, 17, ?	8

Q26. Practice variant 4: Each letter is shifted forward by 1. How is CAT coded?	8
Q27. Practice variant 4: A person faces North and turns right 3 time(s), each by 90 degrees. Which direction is faced?	8
Q28. Practice variant 4: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	8
Q29. Practice variant 4: All poets are readers. No reader is silent. Which conclusion follows?	8
Q30. Practice variant 4: How many ordered selections of 2 people can be made from 6 distinct people?	8
Q31. Practice variant 4: A bag has 5 red and 3 blue balls. One is drawn at random. Probability of red?	8
Q32. Practice variant 4: A table lists three values: 23, 33, 53. What is their total?	8
Remedial and timed set - Questions 33-48	8
Q33. Practice variant 5: Complete the series: 6, 8, 12, 18, ?	8
Q34. Practice variant 5: Each letter is shifted forward by 2. How is CAT coded?	8
Q35. Practice variant 5: A person faces North and turns right 0 time(s), each by 90 degrees. Which direction is faced?	8
Q36. Practice variant 5: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	8
Q37. Practice variant 5: All poets are readers. No reader is silent. Which conclusion follows?	8
Q38. Practice variant 5: How many ordered selections of 2 people can be made from 5 distinct people?	9
Q39. Practice variant 5: A bag has 6 red and 3 blue balls. One is drawn at random. Probability of red?	9
Q40. Practice variant 5: A table lists three values: 24, 34, 54. What is their total?	9
Q41. Practice variant 6: Complete the series: 7, 9, 13, 19, ?	9
Q42. Practice variant 6: Each letter is shifted forward by 3. How is CAT coded?	9
Q43. Practice variant 6: A person faces North and turns right 1 time(s), each by 90 degrees. Which direction is faced?	9
Q44. Practice variant 6: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	9
Q45. Practice variant 6: All poets are readers. No reader is silent. Which conclusion follows?	9
Q46. Practice variant 6: How many ordered selections of 2 people can be made from 6 distinct people?	9
Q47. Practice variant 6: A bag has 7 red and 3 blue balls. One is drawn at random. Probability of red?	9
Q48. Practice variant 6: A table lists three values: 25, 35, 55. What is their total?	9

PYQS AND ANSWER PRACTICE 9

Complete step-by-step solutions	9
Q1. Practice variant 1: Complete the series: 2, 4, 8, 14, ?	9
Q2. Practice variant 1: Each letter is shifted forward by 1. How is CAT coded?	10
Q3. Practice variant 1: A person faces North and turns right 0 time(s), each by 90 degrees. Which direction is faced?	10
Q4. Practice variant 1: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	10
Q5. Practice variant 1: All poets are readers. No reader is silent. Which conclusion follows?	10
Q6. Practice variant 1: How many ordered selections of 2 people can be made from 5 distinct people?	10

Q7. Practice variant 1: A bag has 2 red and 3 blue balls. One is drawn at random. Probability of red?	10
Q8. Practice variant 1: A table lists three values: 20, 30, 50. What is their total?	10
Q9. Practice variant 2: Complete the series: 3, 5, 9, 15, ?	10
Q10. Practice variant 2: Each letter is shifted forward by 2. How is CAT coded?	10
Q11. Practice variant 2: A person faces North and turns right 1 time(s), each by 90 degrees. Which direction is faced?	10
Q12. Practice variant 2: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	11
Q13. Practice variant 2: All poets are readers. No reader is silent. Which conclusion follows?	11
Q14. Practice variant 2: How many ordered selections of 2 people can be made from 6 distinct people?	11
Q15. Practice variant 2: A bag has 3 red and 3 blue balls. One is drawn at random. Probability of red?	11
Q16. Practice variant 2: A table lists three values: 21, 31, 51. What is their total?	11
Q17. Practice variant 3: Complete the series: 4, 6, 10, 16, ?	11
Q18. Practice variant 3: Each letter is shifted forward by 3. How is CAT coded?	11
Q19. Practice variant 3: A person faces North and turns right 2 time(s), each by 90 degrees. Which direction is faced?	11
Q20. Practice variant 3: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	11
Q21. Practice variant 3: All poets are readers. No reader is silent. Which conclusion follows?	11
Q22. Practice variant 3: How many ordered selections of 2 people can be made from 5 distinct people?	12
Q23. Practice variant 3: A bag has 4 red and 3 blue balls. One is drawn at random. Probability of red?	12
Q24. Practice variant 3: A table lists three values: 22, 32, 52. What is their total?	12
Q25. Practice variant 4: Complete the series: 5, 7, 11, 17, ?	12
Q26. Practice variant 4: Each letter is shifted forward by 1. How is CAT coded?	12
Q27. Practice variant 4: A person faces North and turns right 3 time(s), each by 90 degrees. Which direction is faced?	12
Q28. Practice variant 4: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	12
Q29. Practice variant 4: All poets are readers. No reader is silent. Which conclusion follows?	12
Q30. Practice variant 4: How many ordered selections of 2 people can be made from 6 distinct people?	12
Q31. Practice variant 4: A bag has 5 red and 3 blue balls. One is drawn at random. Probability of red?	12
Q32. Practice variant 4: A table lists three values: 23, 33, 53. What is their total?	13
Q33. Practice variant 5: Complete the series: 6, 8, 12, 18, ?	13
Q34. Practice variant 5: Each letter is shifted forward by 2. How is CAT coded?	13
Q35. Practice variant 5: A person faces North and turns right 0 time(s), each by 90 degrees. Which direction is faced?	13
Q36. Practice variant 5: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	13
Q37. Practice variant 5: All poets are readers. No reader is silent. Which conclusion follows?	13
Q38. Practice variant 5: How many ordered selections of 2 people can be made from 5 distinct people?	13

Q39. Practice variant 5: A bag has 6 red and 3 blue balls. One is drawn at random. Probability of red?	13
Q40. Practice variant 5: A table lists three values: 24, 34, 54. What is their total?	13
Q41. Practice variant 6: Complete the series: 7, 9, 13, 19, ?	13
Q42. Practice variant 6: Each letter is shifted forward by 3. How is CAT coded?	14
Q43. Practice variant 6: A person faces North and turns right 1 time(s), each by 90 degrees. Which direction is faced?	14
Q44. Practice variant 6: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?	14
Q45. Practice variant 6: All poets are readers. No reader is silent. Which conclusion follows?	14
Q46. Practice variant 6: How many ordered selections of 2 people can be made from 6 distinct people?	14
Q47. Practice variant 6: A bag has 7 red and 3 blue balls. One is drawn at random. Probability of red?	14
Q48. Practice variant 6: A table lists three values: 25, 35, 55. What is their total?	14
Verified PYQ routing ledger	14
Final error-log checklist	16

Logical Reasoning Coding Counting and DI - Solved Practice Workbook

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BASIC MCQS / REMEDIATION

Diagnostic set - Questions 1-16

Q1. Practice variant 1: Complete the series: 2, 4, 8, 14, ?

A. 22 B. 23 C. 21 D. 24

Q2. Practice variant 1: Each letter is shifted forward by 1. How is CAT coded?

A. CAT B. DBU C. ECV D. None of these (1)

Q3. Practice variant 1: A person faces North and turns right 0 time(s), each by 90 degrees. Which direction is faced?

A. East B. South C. North D. West

Q4. Practice variant 1: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Q5. Practice variant 1: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Q6. Practice variant 1: How many ordered selections of 2 people can be made from 5 distinct people?

A. 21 B. 20 C. 19 D. 22

Q7. Practice variant 1: A bag has 2 red and 3 blue balls. One is drawn at random. Probability of red?

A. $\frac{3}{5}$ B. $\frac{1}{2}$ C. $\frac{2}{5}$ D. 1

Q8. Practice variant 1: A table lists three values: 20, 30, 50. What is their total?

A. 101 B. 99 C. 102 D. 100

Q9. Practice variant 2: Complete the series: 3, 5, 9, 15, ?

A. 23 B. 24 C. 22 D. 25

Q10. Practice variant 2: Each letter is shifted forward by 2. How is CAT coded?

A. DBU B. ECV C. CAT D. None of these (1)

Q11. Practice variant 2: A person faces North and turns right 1 time(s), each by 90 degrees. Which direction is faced?

A. North B. South C. East D. West

Q12. Practice variant 2: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Q13. Practice variant 2: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Q14. Practice variant 2: How many ordered selections of 2 people can be made from 6 distinct people?

A. 31 B. 30 C. 29 D. 32

Q15. Practice variant 2: A bag has 3 red and 3 blue balls. One is drawn at random. Probability of red?

A. $\frac{3}{6}$ B. 1 C. $\frac{1}{2}$ D. None of these (1)

Q16. Practice variant 2: A table lists three values: 21, 31, 51. What is their total?

A. 104 B. 102 C. 105 D. 103

Graded set - Questions 17-32

Q17. Practice variant 3: Complete the series: 4, 6, 10, 16, ?

A. 24 B. 25 C. 23 D. 26

Q18. Practice variant 3: Each letter is shifted forward by 3. How is CAT coded?

A. DBU B. FDW C. CAT D. ECV

Q19. Practice variant 3: A person faces North and turns right 2 time(s), each by 90 degrees. Which direction is faced?

A. North B. East C. South D. West

Q20. Practice variant 3: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Q21. Practice variant 3: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Q22. Practice variant 3: How many ordered selections of 2 people can be made from 5 distinct people?

A. 21 B. 20 C. 19 D. 22

Q23. Practice variant 3: A bag has 4 red and 3 blue balls. One is drawn at random. Probability of red?

A. $\frac{3}{7}$ B. $\frac{1}{2}$ C. $\frac{4}{7}$ D. 1

Q24. Practice variant 3: A table lists three values: 22, 32, 52. What is their total?

A. 107 B. 105 C. 108 D. 106

Q25. Practice variant 4: Complete the series: 5, 7, 11, 17, ?

A. 25 B. 26 C. 24 D. 27

Q26. Practice variant 4: Each letter is shifted forward by 1. How is CAT coded?

A. CAT B. DBU C. ECV D. None of these (1)

Q27. Practice variant 4: A person faces North and turns right 3 time(s), each by 90 degrees. Which direction is faced?

A. North B. East C. West D. South

Q28. Practice variant 4: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Q29. Practice variant 4: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Q30. Practice variant 4: How many ordered selections of 2 people can be made from 6 distinct people?

A. 31 B. 30 C. 29 D. 32

Q31. Practice variant 4: A bag has 5 red and 3 blue balls. One is drawn at random. Probability of red?

A. $\frac{3}{8}$ B. $\frac{1}{2}$ C. $\frac{5}{8}$ D. 1

Q32. Practice variant 4: A table lists three values: 23, 33, 53. What is their total?

A. 110 B. 108 C. 111 D. 109

Remedial and timed set - Questions 33-48

Q33. Practice variant 5: Complete the series: 6, 8, 12, 18, ?

A. 26 B. 27 C. 25 D. 28

Q34. Practice variant 5: Each letter is shifted forward by 2. How is CAT coded?

A. DBU B. ECV C. CAT D. None of these (1)

Q35. Practice variant 5: A person faces North and turns right 0 time(s), each by 90 degrees. Which direction is faced?

A. East B. South C. North D. West

Q36. Practice variant 5: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Q37. Practice variant 5: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Q38. Practice variant 5: How many ordered selections of 2 people can be made from 5 distinct people?

A. 21 B. 20 C. 19 D. 22

Q39. Practice variant 5: A bag has 6 red and 3 blue balls. One is drawn at random. Probability of red?

A. $\frac{3}{9}$ B. $\frac{1}{2}$ C. $\frac{2}{3}$ D. 1

Q40. Practice variant 5: A table lists three values: 24, 34, 54. What is their total?

A. 113 B. 111 C. 114 D. 112

Q41. Practice variant 6: Complete the series: 7, 9, 13, 19, ?

A. 27 B. 28 C. 26 D. 29

Q42. Practice variant 6: Each letter is shifted forward by 3. How is CAT coded?

A. DBU B. FDW C. CAT D. ECV

Q43. Practice variant 6: A person faces North and turns right 1 time(s), each by 90 degrees. Which direction is faced?

A. North B. South C. East D. West

Q44. Practice variant 6: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Q45. Practice variant 6: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Q46. Practice variant 6: How many ordered selections of 2 people can be made from 6 distinct people?

A. 31 B. 30 C. 29 D. 32

Q47. Practice variant 6: A bag has 7 red and 3 blue balls. One is drawn at random. Probability of red?

A. $\frac{3}{10}$ B. $\frac{1}{2}$ C. $\frac{7}{10}$ D. 1

Q48. Practice variant 6: A table lists three values: 25, 35, 55. What is their total?

A. 116 B. 114 C. 117 D. 115

PYQS AND ANSWER PRACTICE

Complete step-by-step solutions

Q1. Practice variant 1: Complete the series: 2, 4, 8, 14, ?

A. 22 B. 23 C. 21 D. 24

Correct answer: A. The successive increments are 2, 4, 6, 8.

Q2. Practice variant 1: Each letter is shifted forward by 1. How is CAT coded?

A. CAT B. DBU C. ECV D. None of these (1)

Correct answer: B. Apply the same alphabetic shift to every letter and reverse-check.

Q3. Practice variant 1: A person faces North and turns right 0 time(s), each by 90 degrees. Which direction is faced?

A. East B. South C. North D. West

Correct answer: C. Represent orientation on a four-direction cycle.

Q4. Practice variant 1: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Correct answer: D. Mohan's son is Riya's parent, so Riya is Mohan's granddaughter.

Q5. Practice variant 1: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Correct answer: A. Poets are inside readers, and readers do not overlap silent people.

Q6. Practice variant 1: How many ordered selections of 2 people can be made from 5 distinct people?

A. 21 B. 20 C. 19 D. 22

Correct answer: B. Order matters, so use nPr.

Q7. Practice variant 1: A bag has 2 red and 3 blue balls. One is drawn at random. Probability of red?

A. $\frac{3}{5}$ B. $\frac{1}{2}$ C. $\frac{2}{5}$ D. 1

Correct answer: C. Favourable equally likely outcomes divided by total outcomes.

Q8. Practice variant 1: A table lists three values: 20, 30, 50. What is their total?

A. 101 B. 99 C. 102 D. 100

Correct answer: D. Read the same unit and add all rows once.

Q9. Practice variant 2: Complete the series: 3, 5, 9, 15, ?

A. 23 B. 24 C. 22 D. 25

Correct answer: A. The successive increments are 2, 4, 6, 8.

Q10. Practice variant 2: Each letter is shifted forward by 2. How is CAT coded?

A. DBU B. ECV C. CAT D. None of these (1)

Correct answer: B. Apply the same alphabetic shift to every letter and reverse-check.

Q11. Practice variant 2: A person faces North and turns right 1 time(s), each by 90 degrees. Which direction is faced?

A. North B. South C. East D. West

Correct answer: C. Represent orientation on a four-direction cycle.

Q12. Practice variant 2: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Correct answer: D. Mohan's son is Riya's parent, so Riya is Mohan's granddaughter.

Q13. Practice variant 2: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Correct answer: A. Poets are inside readers, and readers do not overlap silent people.

Q14. Practice variant 2: How many ordered selections of 2 people can be made from 6 distinct people?

A. 31 B. 30 C. 29 D. 32

Correct answer: B. Order matters, so use nPr.

Q15. Practice variant 2: A bag has 3 red and 3 blue balls. One is drawn at random. Probability of red?

A. 3/6 B. 1 C. 1/2 D. None of these (1)

Correct answer: C. Favourable equally likely outcomes divided by total outcomes.

Q16. Practice variant 2: A table lists three values: 21, 31, 51. What is their total?

A. 104 B. 102 C. 105 D. 103

Correct answer: D. Read the same unit and add all rows once.

Q17. Practice variant 3: Complete the series: 4, 6, 10, 16, ?

A. 24 B. 25 C. 23 D. 26

Correct answer: A. The successive increments are 2, 4, 6, 8.

Q18. Practice variant 3: Each letter is shifted forward by 3. How is CAT coded?

A. DBU B. FDW C. CAT D. ECV

Correct answer: B. Apply the same alphabetic shift to every letter and reverse-check.

Q19. Practice variant 3: A person faces North and turns right 2 time(s), each by 90 degrees. Which direction is faced?

A. North B. East C. South D. West

Correct answer: C. Represent orientation on a four-direction cycle.

Q20. Practice variant 3: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Correct answer: D. Mohan's son is Riya's parent, so Riya is Mohan's granddaughter.

Q21. Practice variant 3: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Correct answer: A. Poets are inside readers, and readers do not overlap silent people.

Q22. Practice variant 3: How many ordered selections of 2 people can be made from 5 distinct people?

A. 21 B. 20 C. 19 D. 22

Correct answer: B. Order matters, so use nPr.

Q23. Practice variant 3: A bag has 4 red and 3 blue balls. One is drawn at random. Probability of red?

A. 3/7 B. 1/2 C. 4/7 D. 1

Correct answer: C. Favourable equally likely outcomes divided by total outcomes.

Q24. Practice variant 3: A table lists three values: 22, 32, 52. What is their total?

A. 107 B. 105 C. 108 D. 106

Correct answer: D. Read the same unit and add all rows once.

Q25. Practice variant 4: Complete the series: 5, 7, 11, 17, ?

A. 25 B. 26 C. 24 D. 27

Correct answer: A. The successive increments are 2, 4, 6, 8.

Q26. Practice variant 4: Each letter is shifted forward by 1. How is CAT coded?

A. CAT B. DBU C. ECV D. None of these (1)

Correct answer: B. Apply the same alphabetic shift to every letter and reverse-check.

Q27. Practice variant 4: A person faces North and turns right 3 time(s), each by 90 degrees. Which direction is faced?

A. North B. East C. West D. South

Correct answer: C. Represent orientation on a four-direction cycle.

Q28. Practice variant 4: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Correct answer: D. Mohan's son is Riya's parent, so Riya is Mohan's granddaughter.

Q29. Practice variant 4: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Correct answer: A. Poets are inside readers, and readers do not overlap silent people.

Q30. Practice variant 4: How many ordered selections of 2 people can be made from 6 distinct people?

A. 31 B. 30 C. 29 D. 32

Correct answer: B. Order matters, so use nPr.

Q31. Practice variant 4: A bag has 5 red and 3 blue balls. One is drawn at random. Probability of red?

A. 3/8 B. 1/2 C. 5/8 D. 1

Correct answer: C. Favourable equally likely outcomes divided by total outcomes.

Q32. Practice variant 4: A table lists three values: 23, 33, 53. What is their total?

A. 110 B. 108 C. 111 D. 109

Correct answer: D. Read the same unit and add all rows once.

Q33. Practice variant 5: Complete the series: 6, 8, 12, 18, ?

A. 26 B. 27 C. 25 D. 28

Correct answer: A. The successive increments are 2, 4, 6, 8.

Q34. Practice variant 5: Each letter is shifted forward by 2. How is CAT coded?

A. DBU B. ECV C. CAT D. None of these (1)

Correct answer: B. Apply the same alphabetic shift to every letter and reverse-check.

Q35. Practice variant 5: A person faces North and turns right 0 time(s), each by 90 degrees. Which direction is faced?

A. East B. South C. North D. West

Correct answer: C. Represent orientation on a four-direction cycle.

Q36. Practice variant 5: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Correct answer: D. Mohan's son is Riya's parent, so Riya is Mohan's granddaughter.

Q37. Practice variant 5: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Correct answer: A. Poets are inside readers, and readers do not overlap silent people.

Q38. Practice variant 5: How many ordered selections of 2 people can be made from 5 distinct people?

A. 21 B. 20 C. 19 D. 22

Correct answer: B. Order matters, so use nPr .

Q39. Practice variant 5: A bag has 6 red and 3 blue balls. One is drawn at random. Probability of red?

A. $\frac{3}{9}$ B. $\frac{1}{2}$ C. $\frac{2}{3}$ D. 1

Correct answer: C. Favourable equally likely outcomes divided by total outcomes.

Q40. Practice variant 5: A table lists three values: 24, 34, 54. What is their total?

A. 113 B. 111 C. 114 D. 112

Correct answer: D. Read the same unit and add all rows once.

Q41. Practice variant 6: Complete the series: 7, 9, 13, 19, ?

A. 27 B. 28 C. 26 D. 29

Correct answer: A. The successive increments are 2, 4, 6, 8.

Q42. Practice variant 6: Each letter is shifted forward by 3. How is CAT coded?

A. DBU B. FDW C. CAT D. ECV

Correct answer: B. Apply the same alphabetic shift to every letter and reverse-check.

Q43. Practice variant 6: A person faces North and turns right 1 time(s), each by 90 degrees. Which direction is faced?

A. North B. South C. East D. West

Correct answer: C. Represent orientation on a four-direction cycle.

Q44. Practice variant 6: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Correct answer: D. Mohan's son is Riya's parent, so Riya is Mohan's granddaughter.

Q45. Practice variant 6: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Correct answer: A. Poets are inside readers, and readers do not overlap silent people.

Q46. Practice variant 6: How many ordered selections of 2 people can be made from 6 distinct people?

A. 31 B. 30 C. 29 D. 32

Correct answer: B. Order matters, so use nPr .

Q47. Practice variant 6: A bag has 7 red and 3 blue balls. One is drawn at random. Probability of red?

A. $3/10$ B. $1/2$ C. $7/10$ D. 1

Correct answer: C. Favourable equally likely outcomes divided by total outcomes.

Q48. Practice variant 6: A table lists three values: 25, 35, 55. What is their total?

A. 116 B. 114 C. 117 D. 115

Correct answer: D. Read the same unit and add all rows once.

Verified PYQ routing ledger

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2024	24	Sequence completion	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	45	Digit-frequency counting	B (supplied)	Use place value, cyclicity or a constrained enumeration.
2024	48	Alphabetic addition	A (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	49	Constrained-triplet count	D (supplied)	Use the total crossing length.
2024	56	Multi-turn direction sense	C (supplied)	Use coordinates and final displacement.
2024	57	Rotational direction sense	D (supplied)	Use coordinates and final displacement.
2024	58	Conclusion validity	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2024	74	Implication consistency	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	75	Operator substitution	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	77	Alphabetic-number coding	A (supplied)	Infer one rule that fits every example and reverse-check.
2024	78	Letter-rearrangement coding	D (supplied)	Place hard constraints in a grid before testing options.
2024	80	Coded-value constraints	B (supplied)	Use the total crossing length.
2025	8	Month-pattern sequence	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	29	Alphabetic subtraction	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	30	Letter-pattern completion	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	38	Coded-operator arrangement	B (supplied)	Place hard constraints in a grid before testing options.
2025	49	Suspect-statement logic	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	50	Tournament-score logic	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	55	Family-relation deductions	C (supplied)	Draw a labelled generation graph.
2025	56	Multiplicative-letter code	A (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	57	Letter-digit substitution	D (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	58	Constrained-digit arrangement	B (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	65	Direction-network distance	C (supplied)	Add rates and invert the net rate.
2025	66	Numeric-cube code	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	9	Constrained-word permutation count	D (provisional)	Use the total crossing length.
2026	14	Circular-seating arrangement	A (provisional)	Place hard constraints in a grid before testing options.
2026	15	Painted-region pattern count	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	23	Weighing-combination count	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2026	33	Ranking-order logic	D (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	34	Segment-matching puzzle	C (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	35	Group-seating constraints	B (provisional)	Use the total crossing length.
2026	36	Statement-implication logic	A (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	44	Blood-relation tree	D (provisional)	Draw a labelled generation graph.
2026	48	Transitive-colour syllogism	D (provisional)	Use set containment and reject an illicit converse.
2026	51	Odd-item weighing logic	A (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	56	Direction turns and distance	D (provisional)	Use coordinates and final displacement.
2026	67	Letter-code decoding	B (provisional)	Infer one rule that fits every example and reverse-check.
2026	70	Weighted-score data table	A (provisional)	Read units and denominators before aggregation.
2026	71	Statement-relationship logic	D (provisional)	Draw a labelled generation graph.
2026	72	Communication: relationship matching	B (provisional)	Draw a labelled generation graph.
2026	73	Communication: tool-purpose matching	C (provisional)	Diagnose the barrier or purpose, then choose the feasible inclusive response.
2026	74	Communication: conflict scenario	C (provisional)	Diagnose the barrier or purpose, then choose the feasible inclusive response.
2026	75	Communication: barrier-example matching	D (provisional)	Diagnose the barrier or purpose, then choose the feasible inclusive response.
2026	76	Communication: questionnaire scenario	C (provisional)	Diagnose the barrier or purpose, then choose the feasible inclusive response.
2026	77	Communication: statements	D (provisional)	Diagnose the barrier or purpose, then choose the feasible inclusive response.

Final error-log checklist

- Was the family classified correctly?
- Were units, domains, quantifiers and constraints copied correctly?
- Was the first-principles relation written before a shortcut?
- Was the selected option independently recomputed or supported by the passage?
- Was the mistake logged and followed by a fresh remedial item?